

EMO Logical Pipeline Architecture & Two-Branch Execution Model

• Decoupled GRL • Relational Graph Algebra • Zero Cross-Worker Training Traffic

Branch A: Full-Graph Propagation & Classification (Primary Path)

Phase 0 : Graph Ingestion

- OGB / DGL raw data load
- Undirected edge symmetrization
- Materialize nodes, edges, masks

Phase 3.7 : SIGN Feature Propagation

- K-hop mean-pooling via Spark SQL
- Summarizer.mean JVM vector ops
- Materialize cached hop tables $[x^0 \dots x^k]$

Phase 3.8 : Global Edge-Free Classifier

- LogisticRegression / MLP probe
- Fits on concatenated features
- 100% full-graph node coverage

Phase 5 : Evaluation & Reporting

- Multi-dataset metric aggregation
- Test accuracy / AUC evaluation
- Automated LaTeX & plot generation

Branch B: Community-Decoupled GNNs & CAAN Super-Node Path

Phase 1 : Community Detection

- Spark native LPA (scalable)
- igraph Louvain (driver memory)
- Materialize communities/{alg}/

Phase 2 : Relational Subgraph Extraction

- Boundary tag: $\text{deg_orig} > \text{deg_intra}$
- Intra-community edge filtering
- repartition('community_id') S3 locality

Phase 3 : Decoupled Community GNNs

- Independent GraphSAGE / GATv2
- Trained in executor worker memory
- Zero cross-worker network traffic

Phase 3b : CAAN Super-Node Recovery

- Major community centroid super-nodes
- Broadcast (S, M) auxiliary context
- Recovers 97.8% boundary accuracy

Shared Delta Lake Transactional Storage Plane on Amazon S3

ACID Isolation • Checkpoint Reuse via `_delta_log` • Schema Enforcement • Parquet Format

Delta Lake Tables & Partition Contracts (S3)

URI: `s3://us-east-1-s3-gnn/delta-data/{dataset}/`

Base Graph (P0)

nodes/
edges/
masks/

id, label, features
src, dst
id, split

Δ Delta Parquet

Communities (P1)

communities/
{alg}/

id: LONG
community_id: LONG

Δ Delta Parquet

Subgraphs (P2)

phase2_nodes/
phase2_edges/

id, features, is_boundary
src, dst, comm_id

Δ Delta Parquet

Propagated (P3.7)

hop_k/
features_kk/

hop 1..K mean cache
 $[x^0 || x^1 || \dots || x^k]$

Δ Delta Parquet

Core Technical Contributions

C1: Transactional Lakehouse Storage

ACID experiment isolation, phase checkpoint reuse via `_delta_log`, schema enforcement.

C2: Relational Graph Operations

Boundary detection & CAAN auxiliary super-node construction via Spark SQL joins/aggregations.

C3: Cloud-Native Systems Hardening

Dynamic EBS redirection (F2), YARN overhead sizing (F1), and Arrow zero-copy serialization.